

Self Service Coffee Kiosk Architecture Justification

Lab 3 - Component Modelling and Architectural Pattern Selection

Architecture Selection

We chose Layered Architecture for the Self-Service Coffee Kiosk System. The design separates the touch-screen presentation layer, the order and payment business layer, hardware integration, and the menu data layer. This keeps the kiosk flow understandable while giving each component a focused responsibility.

Reason 1 - Clear separation for a small, fixed workflow

The kiosk has a stable ordering path: a customer selects one of three coffee types, chooses Small or Large, pays by credit card, and receives a printed receipt. A layered structure matches this sequence naturally. The Kiosk Touch UI captures selections, the Order Manager validates them and calculates the total, and the supporting services perform payment, printing, and menu retrieval. The result is easier to test and maintain than a more operationally complex microservices solution.

Reason 2 - Reliable integration with local hardware and data

The system must interact with a touch screen and a receipt printer while reading menu prices. The Receipt Printer Adapter isolates hardware-specific driver commands from ordering logic, and the Menu Repository isolates data queries. If the printer model or menu storage changes, the affected adapter or repository can be replaced without changing the user interface or payment flow.

Security Advantage

The Payment Service is the only component that communicates with the credit-card gateway. The Order Manager sends a payment request through the IPaymentProcessing interface and receives only an authorization result; card details are tokenized and are not stored in the UI, order record, or menu database. Limiting the sensitive integration boundary reduces exposure of payment data and makes access control and audit logging easier to enforce.

Performance Benefit

Layered processing keeps common kiosk actions local and predictable. Menu data can be read through a cached, read-only interface, so customers see coffee options and prices without a remote service call for every screen refresh. The Order Manager coordinates only the necessary payment and print operations after confirmation, reducing unnecessary work and preserving responsive touch-screen interaction during busy periods.